Comparing LQR and PID Controllers for Quadcopter Control Effectiveness and Cost Analysis

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***Abstract*— Quadcopters, a type of Unmanned Arial Vehicle, have gained a major area of interest due to their various applications, small size, low cost, and maneuverability. The flight principle of a quadcopter regulates its thrust and moments to control movements. This paper investigates linear controllers for non-linear systems based on linearized systems, comparing PID and LQR with integral control, examining input/output disturbance effects, and comparing cost and performance. The results of this investigation show that the LQR controller outperforms the PID controller in controlling quadcopters with satisfactory performance and cost results. It offers a balance between performance and cost, handling constant disturbances efficiently. In the non-linear model, there is a minimal difference in performance and controllers' costs due to the strong coupling between system states and constant disturbances.**

Keywords—Quadcopter, LQR, Integral Control, PID, Cost, Performance

I. INTRODUCTION

Quadcopters are unmanned aerial vehicles (UAVs) used for military, commercial, and recreational purposes. They are popular due to their low cost, ease of use, and versatility. Designing and modeling quadcopters are crucial for various engineering disciplines due to their complex dynamics, interdisciplinary nature, and safety benefits.

Quadcopters, or multirotor helicopters with four rotors, have a history dating back to the early 1900s. The first quadcopters appeared in 1907 with the Gyroplane No. 1 [1], but were unstable and impractical. The development continued in the 1920s in France, but quadcopters gained widespread popularity later [2]. Early designs had complex control systems, but by the late 2000s, advancements in GPS, battery life, and Electronic Speed Controllers made them more popular. Today, studies focus on enhancing quadcopters' autonomy and cooperation, with Raffaello D'Andrea demonstrating a Kinect-controlled quadcopter during a TED Global 2013 talk [3].

Quadcopter flight principle uses four propellers for propulsion, controlling thrust, moments, attitude, and position. Four types of movement, including upward downward, forward backward, leftward rightward, and yaw, compensate for weight.

The main challenge that many researchers have with the quadcopter is that its dynamics are coupled with nonlinear multiple input multiple output (MIMO) and passive dynamics [4]. This complexity is further complicated by the fact that the quadcopter has 6DoF degrees of freedom (three translational and three rotational) with only four independent inputs (rotor speeds). The resulting dynamics are highly nonlinear, especially after accounting for complex aerodynamic effects [5].

In the literature, Quadcopter modeling and design with LQR and PID controllers reveals several studies that have been conducted on this topic. In one study [6], A PID controller with manual adjustment was used to control a quadcopter, demonstrating its effectiveness in stabilizing altitude and attitude. Another study [7] the study introduced full-state feedback and PID control methods for a quadcopter model, demonstrating their functionality with suitable initial parameters. A third study analyzed and tested the performance of an LQR control algorithm for quadrotor helicopters. A fourth study [8] presented a neural network control model for quadcopter stabilization, utilizing a classic PID controller and comparing its results with machine learning controllers. A fifth study [9] built a PD controller for altitude stability and succeeded with a simple quadcopter prototype. Finally, these studies demonstrate the effectiveness of LQR and PID controllers in quadcopter modeling and design, as well as the potential for using machine learning and meta-heuristic algorithms to optimize controller performance. This paper investigates linear controller application to non-linear quadcopter model using PID and LQR with integral control, compares controllers' cost and performance, and examines input and output disturbance effects.

II. QUADCOPTER FLIGHT CONTROL MODEL

In order to represent the drone, we need two coordinate frames: an Inertial Frame (IF) and a Body Frame (BF), as shown in Fig. 1. To know the relationship between the inertial frame and the body frame, we need the rotation matrix R represented in the following equation:

$$R = \begin{bmatrix} c\theta\, c\psi & c\psi\, s\theta\, s\varphi - s\psi\, c\varphi & c\psi\, s\theta\, c\varphi + s\psi\, s\varphi \\ c\theta\, s\psi & s\psi\, s\theta\, s\varphi + c\psi\, c\varphi & s\psi\, s\theta\, c\varphi - c\psi\, s\varphi \\ -s\theta & s\varphi\, c\theta & c\varphi\, c\theta \end{bmatrix} \quad (1)$$

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Considering that $sin\theta = s\theta, sin\varphi = s\varphi, \sin\psi = s\psi$
$cos\theta = c\theta, cos\varphi = c\varphi, \cos\psi = c\psi$

φ Rotation angle around x-axis at inertial frame
θ Rotation angle around y-axis at inertial frame
ψ Rotation angle around z-axis at inertial frame

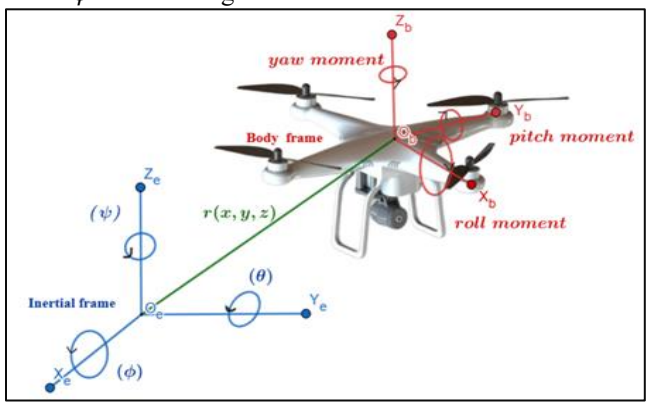


Fig. 1 The representation of the plane on the coordinates of the two frames.

The quadcopter control model consists of four models that illustrated

A. Kinematic model

Kinematic equations assist us in determining the relationship between velocity $\boldsymbol{v}$ and displacement $\boldsymbol{P}$ for the quadcopter's linear and rotational motion [3].

$$\dot{\boldsymbol{P}} = \boldsymbol{v} \tag{2}$$

Considering that $\boldsymbol{P} = [x, y, z]^T$, $\boldsymbol{v} = [\dot{x}, \dot{y}, \dot{z}]^T$

$\boldsymbol{v}$ is the linear velocity of the quadcopter represented in the inertial frame, whereas the linear velocity of the body frame $\boldsymbol{v_b}$ is

$$\boldsymbol{v_b} = R_e^b \, \boldsymbol{v} \tag{3}$$

R_e^b Rotation matrix from body to inertial frame

The following illustration and equation show the relationship between the quadcopter's angular velocity on the body frame and the inertial frame as shown as Fig. 2.

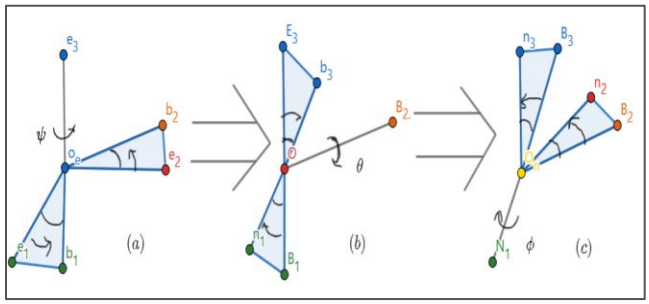


Fig. 2 Euler angles represent rotation matrix in steps: (a) Around Z-axis. (b) Around Y-axis. (c) Around X-axis.

$$\boldsymbol{\omega_b} = \begin{bmatrix} p \\ q \\ r \end{bmatrix} = e_3^b \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + b_2^b \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + n_1^b \begin{bmatrix} \dot{\varphi} \\ 0 \\ 0 \end{bmatrix} \tag{4}$$

e_3^b Frame (a) rotation around e_3 vector
b_2^b Frame (b) rotation around b_2 vector
n_1^b Frame (c) rotation around n_1 vector

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = T_r \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \tag{5}$$

p Angular rate around x-axis at body frame
q Angular rate around x-axis at body frame
r Angular rate around x-axis at body frame

$$T_r = \begin{bmatrix} 1 & 0 & -sin\theta \\ 0 & cos\varphi & sin\varphi . cos\theta \\ 0 & -sin\varphi & cos\varphi . cos\theta \end{bmatrix} \tag{6}$$

Therefore, angular velocities at inertial frame are

$$\dot{\boldsymbol{\Theta}} = \begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = {T_r}^{-1} \boldsymbol{\omega_b} \tag{7}$$

B. Dynamic model

The rigid body dynamics have been separated into two sections for the analysis of quadcopter dynamics: the position dynamic model and the attitude dynamic model, with five assumptions [3]

1) The multicopter is a rigid body.
2) The multicopter is simply powered by gravity and propeller thrust.
3) The geometric center and the Center of Gravity (CoG) of the multicopter are the same.
4) Gyroscopic torques are neglected.
5) The mass and the moments of inertia are constant.

Analyze the quadcopter's forces using Newton's second law, which describes the forces applied to it, as shown in Fig. 3 in the translational dynamics section.

$$m \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix} + R_b^e \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix} \tag{8}$$

m Mass of quadcopter
g Gravity
T Thrust

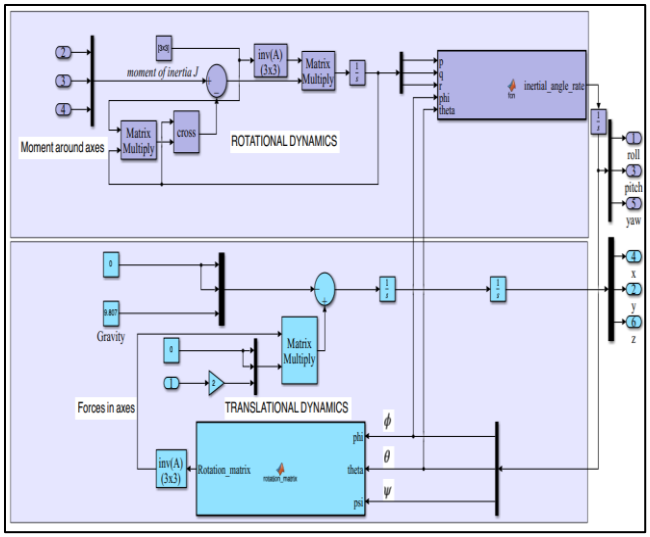


Fig. 3 Nonlinear dynamic model of quadcopter.

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The resultant moments on the quadcopter were computed with the Coriolis Effect included, as presented in Fig. 3. of the rotational dynamics section.

$$J.\dot{\boldsymbol{\omega}}_{\boldsymbol{b}} = \boldsymbol{\tau} - \boldsymbol{\omega}_{\boldsymbol{b}} \times J\boldsymbol{\omega}_{\boldsymbol{b}} \tag{9}$$

$\boldsymbol{\tau}$ Torques around three axes

Moment of inertia matrix J according to the assumption (3)

$$J = \begin{bmatrix} J_{xx} & 0 & 0 \\ 0 & J_{yy} & 0 \\ 0 & 0 & J_{zz} \end{bmatrix} \tag{10}$$

C. Control effectiveness model

In the quadcopter, the actor in its movement is the speed of the four motors, and in order to be able to control it, we link it with the dynamic laws (11) and (12), namely the thrust and the moments around its axes for + configuration [3].

$$\begin{bmatrix} T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} c_T & c_T & c_T & c_T \\ 0 & -dc_T & 0 & dc_T \\ dc_T & 0 & -dc_T & 0 \\ c_M & -c_M & c_M & -c_M \end{bmatrix} \begin{bmatrix} \omega_1^2 \\ \omega_2^2 \\ \omega_3^2 \\ \omega_4^2 \end{bmatrix} \tag{11}$$

ω_i Motor speed c_M Torque coefficient
c_T Thrust coefficient d Arm length

D. Propulsion model

The Electronic Speed Controller (ESC) generates the control signal U_m for motor speed; it is powered by battery U_b and throttle command σ. Throttle command sets power consumption from 0 to 1 as shown as [3]

$$U_m = \sigma U_b \tag{12}$$

The steady state of motor speed ω_{ss} is expressed as [3]

$$\omega_{ss} = C_b U_m + \bar{\omega}_b = C_R \sigma + \bar{\omega}_b \tag{13}$$

$\bar{\omega}_b$ and C_R are constant parameters approximating the relationship between throttle commands and motor speed as defined by Quan Quan [3].

Brushless Direct Current BLDC motor speed ω_i can be modeled as a simplified as first order transfer function [3]

$$\omega_i = \frac{1}{T_m s + 1} \omega_{ss} \tag{14}$$

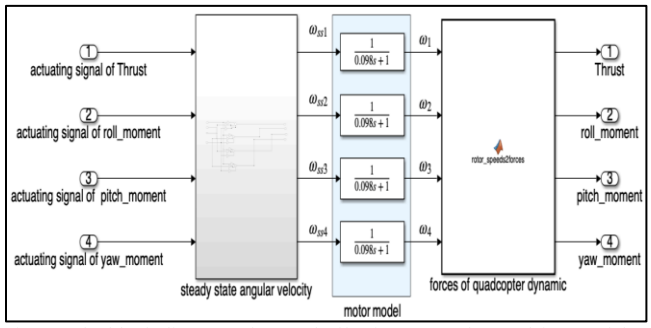


Fig. 4 The block diagram of control effectiveness and propulsion model.

III. LINEARIZATION QUADCOPTER MODEL

To linearize the quadcopter model, the state space representation can be used as the following equations

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u}) \tag{15}$$

Where the states of the system are

$$\boldsymbol{x} = [x, y, z, \varphi, \theta, \psi, u, v, w, p, q, r]^T$$

In addition, inputs of system are

$$\boldsymbol{u} = [T, \tau_x, \tau_y, \tau_z]^T$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \\ \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \\ \dot{u} \\ \dot{v} \\ \dot{w} \\ \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} u \\ v \\ w \\ \mathrm{p} + \tan\theta\, sin\varphi.\, \mathrm{q} + \tan\theta\, cos\varphi\, .r \\ cos\varphi.q - sin\varphi.r \\ \frac{sin\varphi}{cos\theta}.q + \frac{cos\varphi}{cos\theta}.r \\ (cos\psi\, sin\theta\, cos\varphi\, +\, sin\psi\, sin\varphi)\frac{T}{m} \\ (sin\psi\, sin\theta\, cos\varphi\, -\, cos\psi\, sin\varphi)\frac{T}{m} \\ -g + cos\varphi\, cos\theta\frac{T}{m} \\ \frac{1}{J_{xx}}(\tau_x - (J_{zz} - J_{yy})\, qr) \\ \frac{1}{J_{yy}}(\tau_y - (J_{xx} - J_{zz})\, pr) \\ \frac{1}{J_{zz}}(\tau_z - (J_{yy} - J_{xx})\, pq) \end{bmatrix} \tag{16}$$

Therefore, apply the Taylor series at the equilibrium point (the hovering point where the linearized model of the quadcopter can be designed to stabilize it) of the states and inputs of the model, which are

$$\boldsymbol{x_e} = [0,0,0,0,0,0,0,0,0,0,0,0]^T \quad , \boldsymbol{u_e} = [mg, 0,0,0]^T$$

And substitute them into (17)

$$\dot{\vec{\boldsymbol{x}}} = f(\boldsymbol{x_e}, \boldsymbol{u_e}) + \frac{\partial f(\boldsymbol{x},\boldsymbol{u})}{\partial \boldsymbol{x}}\Big|_{\substack{\boldsymbol{x}=\boldsymbol{x_e} \\ \boldsymbol{u}=\boldsymbol{u_e}}} (\vec{\boldsymbol{x}} - \boldsymbol{x_e}) + \frac{\partial f(\boldsymbol{x},\boldsymbol{u})}{\partial \boldsymbol{u}}\Big|_{\substack{\boldsymbol{x}=\boldsymbol{x_e} \\ \boldsymbol{u}=\boldsymbol{u_e}}} (\vec{\boldsymbol{u}} - \boldsymbol{u_e}) + H.O.T \tag{17}$$

Higher Order Term($H.O.T$)

Therefore, linearized state space model is

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + B\boldsymbol{u}\,, \qquad y = C\boldsymbol{x} + D\boldsymbol{u} \tag{18}$$

$$A = \begin{bmatrix} 0_{6*6} & & & & & I_6 \\ 0 & -g & 0 & & & \\ 0_{3*3} & g & 0 & 0 & 0_{6*6} & \\ 0 & 0 & 0 & & & \\ 0_{3*3} & & 0_{3*3} & & & \end{bmatrix} \tag{19}$$

$$B = \begin{bmatrix} & 1/m & 0 & 0 & 0 \\ 0_{4*8} & 0 & 1/J_{xx} & 0 & 0 \\ & 0 & 0 & 1/J_{yy} & 0 \\ & 0 & 0 & 0 & 1/J_{zz} \end{bmatrix}^T \tag{20}$$

After analyzing matrices, A and B, the eigenvalues of the system are zero. This indicates that the system is not stable and requires a controller to achieve stability.

All parameters were taken from Quan [3], except for body mass and inertial mass about the axes, which were taken as approximations. See table 1.

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I. QUADCOPTER PARAMETERS VALUES.

Parameter	*Symbol*	Value	Unit
Mass	m	0.5	kg
Mass of inertia	J_{xx}	8×10^{-3}	$kg.m^2$
Mass of inertia	J_{yy}	8×10^{-3}	$kg.m^2$
Mass of inertia	J_{zz}	15×10^{-3}	$kg.m^2$
Thrust coefficient	c_T	1.984×10^{-7}	N/RPM^2
Torque coefficient	c_M	3.733×10^{-9}	$N.m/RPM^2$
Rotation speed factor	C_b	571.35	RPM/V
Rotation speed offset	$\bar{\omega}_b$	1779	RPM
Arm length	d	0.223	m
Time constant	T_m	0.098	s
Battery voltage	U_b	11.1	V

IV. DESIGN CONTROL SYSTEM

In this paper, the quadcopter system is controlled using Proportional Integral Derivative PID and Linear Quadratic Regulator LQR with Integral control, and the results of both methods are compared.

A. PID implementation

The quadcopter's control system uses four inputs and six outputs to regulate altitude, position, and attitude. A basic PID is sufficient for altitude and yaw, but a cascade PID is needed for attitude and horizontal position. An efficient cascade system consists of two loops, with the inner loop closed for efficient response. The inner loop should respond faster than the outer loop [10].

To tune a quadcopter's control system using PID Tuner tool in SIMULINK: (1) Tune inner loop controllers for faster response; (2) Tune outer loop controllers; and (3) Ensure responsiveness in non-linear system as shown in Table 2.

II. THE TUNED PARAMETER OF PID CONTROLLERS

PID controller name	PID parameters			
	k_p	k_i	k_d	N
Altitude controller	0.45	0.053	0.99	100
position controller	0.004	0.0001	0.036	100
attitude controller	0.015	0.0003	0.135	100
Yaw controller	0.005	0.0005	0.01	100

a. N is filter coeficient.

B. LQR with Integral control implementation

To use LQR with integral control for quadcopter, add augmented states for integral control, then adjust Q and R weights to eliminate controllability problem.

State-space design allows for integral control by adding the desired integral x to the state vector. The differential equation is [11]

$$\dot{x}_A = r - Hx \tag{21}$$

The equation (21) is augmented to state equations of the nominal system as [11]

$$\begin{bmatrix} \dot{x} \\ \dot{x}_A \end{bmatrix} = \begin{bmatrix} A & 0 \\ -H & 0 \end{bmatrix} \begin{bmatrix} x \\ x_A \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} w_1 \tag{22}$$

$$y = [C \quad 0] \begin{bmatrix} x \\ x_A \end{bmatrix} + [B_2 \quad 0]\, w_2 \tag{23}$$

The H matrix is the augmented states (x, y, z, ψ) to be controlled in the quadcopter control system, and w's are input and output disturbance. The LQR algorithm addresses the challenge of balancing system response with control effort by computing the gain K while minimizing cost function J.

$$\mathcal{J} = \int_0^\infty [\boldsymbol{x^T Q x} + \boldsymbol{u^T R u}]\, dt \tag{24}$$

Designing a controller involves selecting appropriate weights for state and control matrices, Q and R, in a multi-input system. The goal is to balance state errors and effective control input use. Weights are typically chosen through trial and error, with Q being a diagonal matrix with weight factors assigned to state vector elements and R being a diagonal matrix with weight factors assigned to input vector components [11]. The LQR controller requires integral control to eliminate steady-state errors and address controllability issues and narrow tuning ranges. To address these, an augmented state is added for each state, which is then controlled using LQR.

The feedback law after the design of the K matrix by LQR is [11]

$$\boldsymbol{u} = -[k \quad k_A] \begin{bmatrix} \boldsymbol{x} \\ \boldsymbol{x_A} \end{bmatrix}, \qquad K = [k \quad k_A] \tag{25}$$

As a starting point for assigning the weights of the Q and R matrices, we use Bryson's rule, which is a widely used heuristic that provides a good initial guess for the weighting matrices Q and R. So what is Bryson's rule?

The rule is that if [11]

$$y_1 = \alpha m_1, y_2 = \alpha m_2, y_3 = \alpha m_3 \tag{26}$$

Where m's are the maximum acceptable error deviation of n states, and if Q is a diagonal matrix, then

$$q_{11} y_1^2 = q_{22} y_2^2 = q_{33} y_3^2 \tag{27}$$

Thus

$$q_{11} \alpha^2 m_1^2 = q_{22} \alpha^2 m_2^2 = q_{33} \alpha^2 m_3^2 \tag{28}$$

Then the diagonal matrix elements are satisfactorily solved by understanding the accepted dynamic parameters of the quadcopter and applying them to the following equations [11]

$$q_{11} = \frac{1}{m_1^2}, \qquad q_{22} = \frac{1}{m_2^2}, \qquad q_{33} = \frac{1}{m_3^2} \tag{29}$$

Furthermore, the initial values of the R matrix can be set as in the following equation

$$r_{11} = \frac{1}{u_{1max}^2}, \qquad r_{22} = \frac{1}{u_{2max}^2}, \qquad r_{33} = \frac{1}{u_{3max}^2} \tag{30}$$

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u_{imax} is the maximum acceptable input for the quadcopter system. The following tables show the initial values for the Q and R weights. The value of the feedback gain K was obtained by utilizing the lqr function in MATLAB.

III. THE FINAL VALUES OF CONTROL MATRIX WEIGHTS R

i	1	2	3	4
u_i	*Thrust*	*roll moment*	*pitch moment*	*yaw moment*
R_{ii}	0.01	100	100	100

IV. THE FINAL VALUES OF STATE MATRIX WEIGHTS Q

i	1	2	3	4	5	6	7	8
x_i	x	y	z	ϕ	θ	ψ	$\dot{x}$	$\dot{y}$
Q_{ii}	1	1	1	1	1	1	0.01	0.01
i	**9**	**10**	**11**	**12**	**13**	**14**	**15**	**16**
x_i	$\dot{z}$	$\dot{\phi}$	$\dot{\theta}$	$\dot{\psi}$	x_A	y_A	z_A	ψ_A
Q_{ii}	0.01	0.25	0.25	0.25	30	30	70	1

V. THE FINAL VALUES OF FEEDBACK GAIN MATRIX K

	State	K		State	K
Thrust	z	32.7412	**θ moment**	x	0.5427
	$\dot{z}$	5.807		θ	0.7288
	z_A	-83.666		$\dot{x}$	0.2597
ϕ moment	y	-0.5427		$\dot{\theta}$	0.119
	ϕ	0.7288		x_A	-0.5477
	$\dot{y}$	-0.2597	**ψ moment**	ψ	0.1625
	$\dot{\phi}$	0.119		$\dot{\psi}$	0.0863
	y_A	0.5477		ψ_A	0.1

The system's bandwidth increases with Q's overall values, damping increases with Q's term weighting velocity type state elements, and system speed can be increased by increasing weights [11].

V. SIMULATION AND RESULTS

Controllers were used to handling inputs with a change of less than 2 meters per second, achieved through trial and error. (1) It is assumed that linear controllers controlling a linear system have good and close performance, and this is evident in Fig. 5 and Fig. 6.

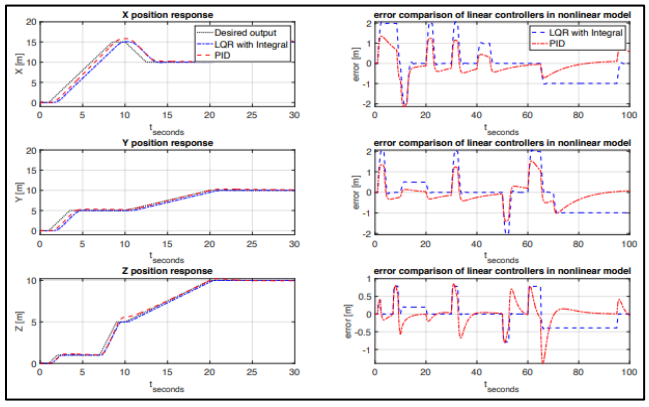


Fig. 5 The trajectory-tracking response to position with error in linearized model.

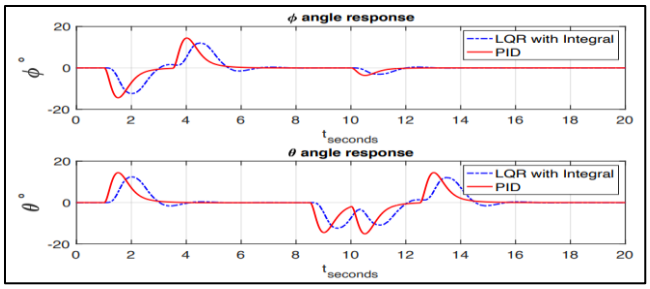


Fig. 6 The effect of trajectory tracking on angles of linearized dynamic model.

(2) When applying controllers to a nonlinear system, a slight difference in performance is expected due to the effect of inclination angles around the horizontal axes of the quadcopter, and this is shown in Figs. 7, 8, and 9.

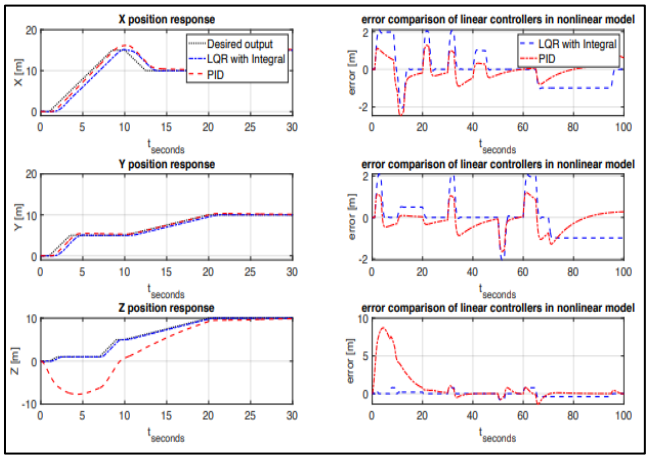


Fig. 7 The trajectory-tracking response to position with error in nonlinear model without disturbance.

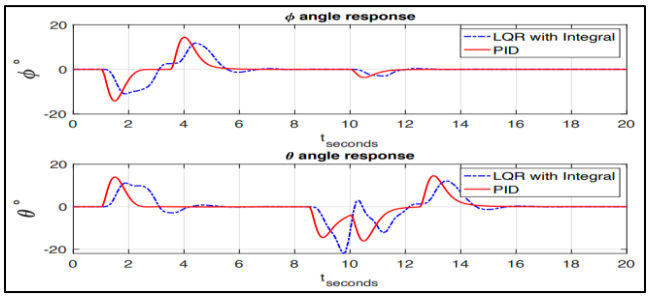


Fig. 8 The effect of trajectory tracking on angles of nonlinear dynamic model without disturbance.

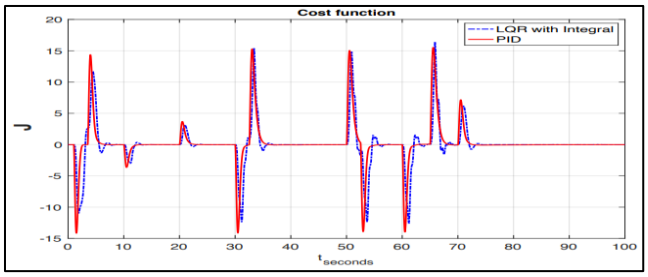


Fig. 9 Cost function of linear controllers for nonlinear model without disturbance.

(3) In comparing controllers for nonlinear models in the presence of disturbances, it was clear that LQR was more effective than PID in quickly removing the effects of disturbances. This is shown in Figs. 10, 11, and 12.

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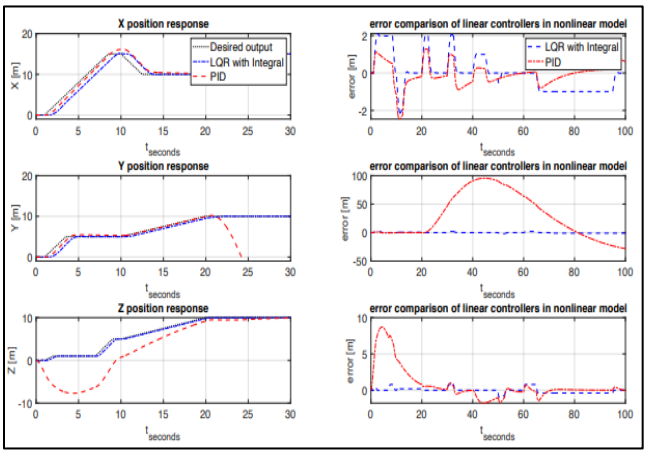


Fig. 10 The trajectory-tracking response to position with error in nonlinear model with disturbance.

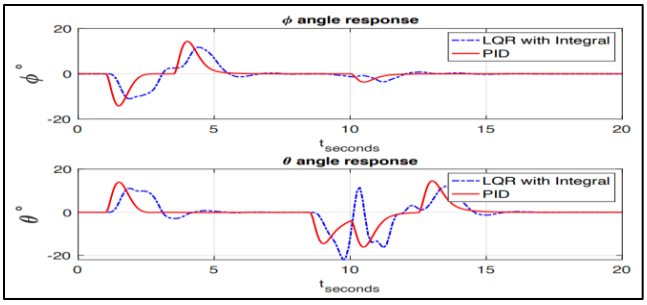


Fig. 11 The effect of trajectory tracking on angles of nonlinear dynamic model with disturbance.

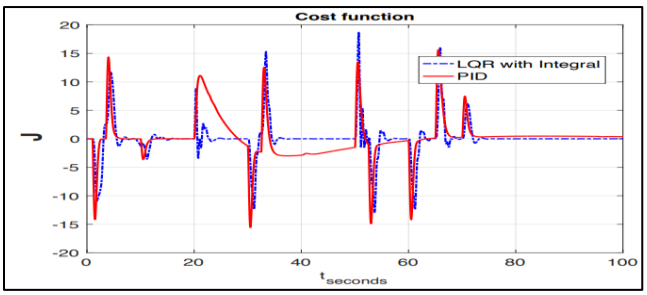


Fig. 12 Cost function of linear controllers for nonlinear model with disturbance.

(4) In the 3D simulation in Fig. 13, the difference between the two controllers is clear in terms of overshoot, where LQR was better than PID.

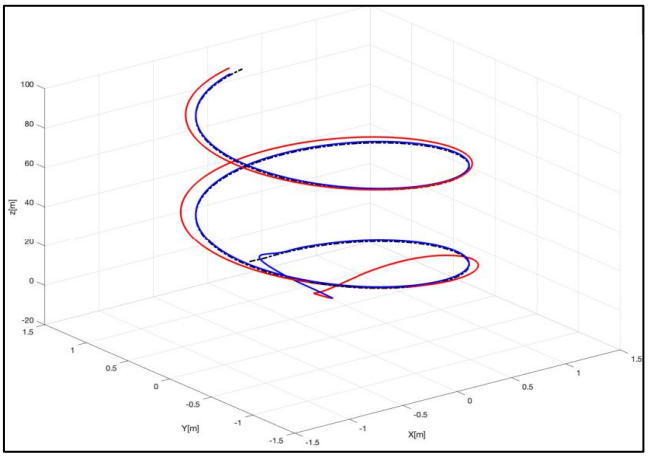


Fig. 13 Comparison of linear controllers in nonlinear model in 3 dimensions.

VI. Conclusion and Future Work

A. Conclusion

PID provided fast tracking but with overshoots, which are undesirable. Steady-state error was good, but PID could not handle disturbances well, making the system unstable. LQR with integral control balanced performance and cost. Tracking was similar to PID with no overshoot and better cost. It could also efficiently handle constant disturbances. When applying controllers to the nonlinear vs linearized models, minimal differences in performance and cost were observed. This was due to strong coupling between nonlinear states. Constant disturbances had a greater influence on the nonlinear model compared to the linearized one.

B. Future work

In the future, it is recommended to use the 2-DoF PID or FOPID (Fractional Order PID) instead of the PID to reduce overshooting. Additionally, a combined control system between LQR and PID can be used to obtain the best results in terms of performance and cost.

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